

ANGIOPLASTIC STENOSIS IN SPONTANEOUS BILATERAL CAROTIC ARTERY DISSECTION: A CASE REPORT

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ABSTRACT

Case: A 28-years-old woman presented with sudden right extremities weakness, right sided facial droop, and not being able to speak. On examination, the patient had upper motor neuron (UMN) hemiplegia and motor aphasia. On blood laboratory, we found D-dimer level of 1068 ng/dl. A non-contrast head CT-scan and CT-angiography showed left cerebral hemisphere infarction, and bilateral carotid artery stenosis. Further investigation with transcranial Doppler demonstrated partial stenosis of internal carotid artery (ICA). On cerebral digital subtraction angiography (DSA), we observed pseudoaneurysm on the right ICA and flame-shaped appearance on the left ICA, both are common radiological finding of ICA dissection. Then we performed stenting angioplasty on bilateral ICA to provide recanalization, and aspirin 80mg q.d. and clopidogrel 75mg q.d. was initiated.

Conclusion: Spontaneous dissection of the carotid artery can occur on young adults. DSA is considered as the gold standard for vascular examination, and stenting with angioplasty was performed for recanalization to prevent the progression to severe stroke.

Keywords: Spontaneous dissection, carotid artery, stenting angioplasty

Introduction

Dissection of internal carotid artery (ICA) is among the most common etiology of ischemic stroke in young adult. Around 10% to 25% of ischemic strokes under the age of 45 occur after ICA dissection. Incidence rate of ICA dissection is around 2.5 to 3.0 per 100,000 individuals and is more common in males (55%).^{1,2}

The etiology of ICA dissection could be either spontaneous, or traumatic. Spontaneous ICA dissection at a young adult often resulted from fibromuscular dysplasia, a genetic connective. Traumatic ICA, in another hand, happened after cervical injury, often caused by sudden neck rotation or cervical manipulation.¹⁻³

Internal carotid artery (ICA) dissection might appear as headache, neck pain, tinnitus, Horner's syndrome, retinal ischemia or hemiparesis.^{4,5} Non-invasive angiography, using computed tomography angiography (CTA) or, even better with magnetic resonance angiography (MRA), could visualize ICA dissection. However, digital subtraction angiography (DSA), an invasive angiography, is the diagnostic gold standard.⁴ Several typical findings of ICA dissection on angiography are stenosis, string signs, intimal flaps, and vessel occlusion.

Case Report

A 28-year-old woman present to our facility with sudden onset of right-sided hemiplegia, right-sided facial droop and not being able to speak when going to sleep at night. There were no other complaints, like headaches, nausea, vomitus, seizures, dizziness and blurred vision. There was no history of hypertension, diabetes and heart disease. Patient's vital signs were within normal limits, and on neurological examination we observed motor aphasia, right sided upper motor neuron (UMN) facial nerve palsy and tetraplegia. Hematological parameters, including hemostasis and electrolytes were unremarkable, except increasing D dimer level of 1068 ng/dl. The patient underwent a plain head computed tomography (CT) scan and we observed hypodensity lesion that indicates infarction in the left cerebral hemisphere. Then, he received aspirin 80 mg q.d.

On transcranial doppler (TCD) we observed partial stenosis of left intracranial ICA and left distal anterior cerebral artery (ACA). Head CT angiography also showed partial stenosis of the left ICA and partial stenosis of the left ACA. On cerebral DSA, we found a flame-shaped artery on the left ICA and a pseudoaneurysm on the right ICA, supporting the diagnosis of bilateral ICA dissection.

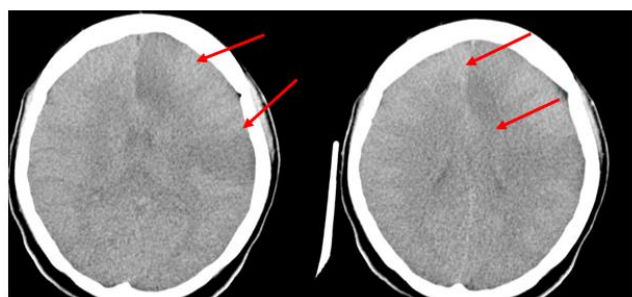


Figure 1. Red arrows show infarction in the left cerebral hemisphere on plain head CT scan

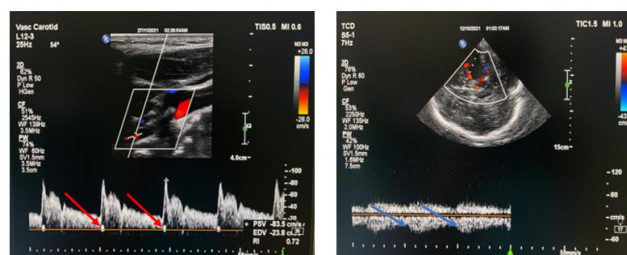


Figure 2. Reverberation of ICA flow (red arrows) and systolic delay (blue arrows) on TCD

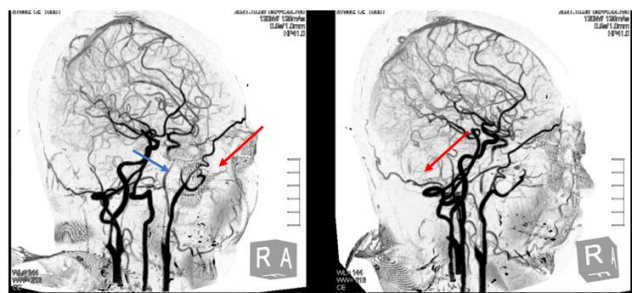


Figure 3. Flamed shape artery on left ICA (red arrows) and pseudoaneurysm on right ICA (blue arrows) on CT angiography

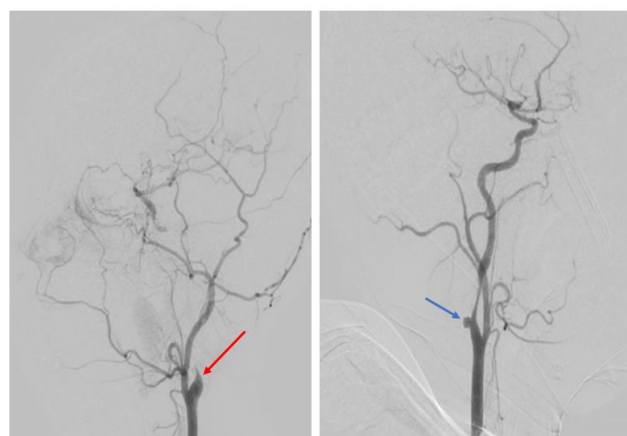


Figure 3. Flamed shape artery on left ICA (red arrows) and pseudoaneurysm on right ICA (blue arrows) on DSA

The patient then underwent stenting angioplasty on bilateral ICA, and followed by double antiplatelet therapy with aspirin 80mg q.d and clopidogrel 75mg q.d. On follow-up session 3 weeks after stenting angiography, patient's complaints of right sides weakness, facial palsy, and dysarthria was improved.

Discussion

Dissection of internal carotid artery (ICA) is among the most common etiology of ischemic stroke in young adult.

Most ICA dissection occur before 50 years old and often happened in man. Arterial dissection is characterized by a tear tunica intima, the innermost layer of the arterial wall and blood penetration that forms an intramural hematoma. The presence of an intramural hematoma results in lumen stenosis, thrombus formation, which could lead to ischemic stroke due to thromboembolism or hypoperfusion.^{1,6,7}

Most arterial dissections occur spontaneously. Spontaneous internal carotid artery dissection can be unilateral or bilateral. But the incidence of bilateral ICA dissection is rare, only on 5% to 28% of cases.⁶

Arterial dissection is frequently associated with disorder of connective tissue, for instance cystic medial necrosis (CMN), fibromuscular dysplasia, and intimal fibroplasia. Among connective tissue diseases, fibromuscular dysplasia is the most common culprit of bilateral ICA dissections. Individuals with fibromuscular dysplasia are relatively young and with no to little cardiovascular risk factors. The arterial segments that are frequently involved in fibromuscular dysplasia are the middle or distal vessels. A novel recurrent COL5A1 genetic variant inherited by autosomal dominant pattern is associated with a dysplasia associated arterial disease exhibiting dissections and fibromuscular dysplasia.^{8,9}

The clinical manifestations of arterial dissection are varied and may not appear for 12 to 24 hours after injury. Spontaneous ICA dissection might be asymptomatic, but are usually preceded by persistent ipsilateral neck pain or headache. The most common clinical signs and symptoms of internal carotid artery dissection include cerebral ischemia (63-90%), headache and/or neck pain (58-92%), and Horner's syndrome (9-75%).^{4,7}

The diagnostic gold standard of ICA dissection is DSA, but other modalities including ultrasound, CT, CTA, magnetic resonance imaging (MRI), and MRA, could help the diagnosis, since there is limited access to DSA. Several radiological features that can be found on angiography are ICA occlusion, narrow and irregular vessel lumens, presence of double lumens, intimal flaps, pseudoaneurysm, or string signs.^{7,12}

Carotid artery dissection prognosis is varied, depending on several factors, stroke severity, and the recurrence rate. However, most of these cases has poor prognosis, resulting in death, or severe permanent neurological deficits. Evaluation with ultrasound modalities unveiled that 50% of cases had complete recanalization after three months, and 60% occurs after six months.¹⁰

There are several modalities of artery dissection, including medical management using anticoagulants or antiplatelets, and non-medical treatment with endovascular intervention. The effectiveness of antiplatelet and anticoagulant therapy in the long-term treatment of secondary stroke investigated in The Cervical Artery Dissection in Stroke Study (CADISS) in 2015, found no significant difference between those two treatments. Treatment duration was 3 to 6 months and INR ranged from 2.0 to 3.0. Several studies suggest the use of stents despite adequate medical therapy. This considers the occurrence of cerebral hypoperfusion and the presence of pseudoaneurysms.¹¹

In our case the patient had sudden onset of right-sided hemiplegia, right-sided facial droop and not being able to

speak, which occurred at a relatively young age. At this age groups, stroke is usually caused by non-atherosclerotic factors, such as anatomical abnormalities of blood vessels or blood vessel dissection.

From the clinical history, physical examination, CT scan, TCD, CTA, and DSA, we conclude that the patient had bilateral ICA dissection. The possible etiology of dissection in our case is connective tissue disorder, which need further investigation.

Conclusion

Dissection of internal carotid artery (ICA) should be considered as the etiology of ischemic stroke in patients 45 years old or less. DSA is the diagnostic gold standard of ICA dissection, and stenting with angioplasty is performed for recanalization of blood vessel, thus prevent the progression to severe stroke.

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Conflict of Interest

The authors declared no conflicts of interest.

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